

Chemist: KR Completed Date: 7/11/25

Checker: TE Review Date: 7/21/2025

IMIS - 1280

Analyte(s) - Particulate fluorides

Sample Type(choose one):

☒ **QCSM**

QC sample numbers: 131048-131063

Data Package Documents & Records

- ☒ A) Initial Standard Screen Print
- ☒ B) Intermediate Standard Screen Print
- ☒ C) Batch Preparation Screen Print
- ☒ D) Batch Preparation Volumes Screen Print
- ☒ E) QCSM Prep Worksheet - Input (each analyte)
- ☒ F) QCSM Prep Worksheet - Spiking (each analyte)
- ☒ G) Certificate of Analysis (for each stock)

☐ **Internal Proficiency/Demonstration of Competence**

Data Package Documents & Records

- ☐ A) Initial Standard Screen Print
- ☐ B) Intermediate Standard Screen Print
- ☐ C) IP Prep Worksheet - Input
- ☐ D) IP Prep Worksheet - Spiking
- ☐ E) Certificate of Analysis (for each stock)
- ☐ F) 91A(s)

☐ **State Exchange: Round**

Data Package Documents & Records

- ☐ A) Initial Standard Screen Print
- ☐ B) Intermediate Standard Screen Print
- ☐ C) Prep Worksheet - Input
- ☐ D) Prep Worksheet - Spiking
- ☐ E) List of Participating Labs
- ☐ F) Letter Sent to Labs/Reporting Sheet
- ☐ G) Certificate of Analysis (for each stock)
- ☐ H) 91A(s)

Other support documents/records (specify):

- ☐ Z1) Grav Verification Printout from LISA
- ☐ Z2) _____
- ☐ Z3) _____

Standard Documentation

- | ✓A | ✓C | |
|-------------------------------------|-------------------------------------|--|
| ☒ | ☒ | 1) Density/Concentration of spiking material |
| ☒ | ☒ | 2) Purity |
| ☒ | ☒ | 3) Manufacturer |
| ☒ | ☒ | 4) Lot Number |
| ☒ | ☒ | 5) Standard ID (if necessary) |
| ☒ | ☒ | 6) Barcode |
| ☒ | ☒ | 7) Date Received/Prepped/Expires |

Media Documentation

- | | | |
|-------------------------------------|-------------------------------------|---------------------|
| ☒ | ☒ | 8) Media Type |
| ☒ | ☒ | 9) Manufacturer |
| ☒ | ☒ | 10) Catalog # |
| ☒ | ☒ | 11) Lot # |
| ☒ | ☒ | 12) Expiration Date |

Sample Labeling

- | | |
|-------------------------------------|--|
| ☒ | 13) Labels attached to sample and packaged/stored correctly |
| ☒ | 14) Sample number on each label matches correct sample number on preparation worksheets |
| ☒ | 15) QCSM Only. Labels attached to outside of box/container identify analyte, IMIS, media |

Documentation Review

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☒ | 16) All hand written entries in indelible ink; corrections use single strikeout (dated & initialed) |
| ☒ | ☒ | 17) Analyst's initials & date on top sheet of associated, or linked, pages of documents |
| ☒ | ☒ | 18) Data exclusion is adequately justified and documented |
| ☒ | ☒ | 19) Documentation produced is sufficient to allow complete re-construction of sample preparation events |

Reagent & Standard Usage

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☒ | 20) Reagents and standards used (spiking solutions & stocks) are <u>not</u> expired |
| ☒ | ☒ | 21) Standard and reagent usage is documented in an <u>explicitly traceable</u> manner (Q-008) |

Data Management Review

- | | | |
|-------------------------------------|-------------------------------------|-----------------------------------|
| ☒ | ☒ | 22) Calculations verified correct |
|-------------------------------------|-------------------------------------|-----------------------------------|

Comments: _____

KR 7/11/25

Initial Standard

Manufacturer Aldrich Chemical Company [Sigma]

Status In-Use

Lot No MKCK9071

Search

Bar Code No 302325A

Search

% Purity 99.1

Date Received 02-JUN-2021

Comments

Physical State Solid

Date
ReevaluatedSodium Fluoride (NaF)
(F mw)/(NaF fw) = 19/42 = 0.4524
Results reported as (F-)

Manuf Titration

Date Expires 02-JUN-2026

SLTC Titration

Analyte Composition

IMIS	Analyte Name	CAS	Part/Density	Unit	Concentration
1460	Hydrogen Fluoride (as F)	7664-39-3	.4524	mg/mL	.448328

Save



OSHA

8
KR 7/11/25

Intermediate Standard

Standard Name	Preparer's Name	First Name	Initial	Update
1280-20240716-KR	Ramji	Karpagavalli	KR	
Prepared Date	Expiration Date	Disposition	State	Re-evaluation Date
16-JUL-2024	16-JUL-2025	In-Use	Solution	
Comment	Made with DI water.			

Parent Standards

MKCK9071

Analyte Composition

IMIS	Analyte Name	Amount	Final Volume	Concentration	Unit
1280	Fluorides (as F)	301.93	15	9.023681	mg/mL

CR 7/11/25

QC Batch

Last Name	Ranji	First Name	Karpagavalli	Initials Id	KR	Media Lot Number	24223-7E8-064
Media Short Desc	MDEF 37	Media Desc	MDEF, 37-mm, 0.8 um, AA37, Millipore MAWP037A0	Comments		Media Expiration Date	30-APR-2029
Prepared On	11-JUL-2025	Comments		Comments_Analyst			
Expiration Interval	12						
Expiration Date	11-JUL-2026						

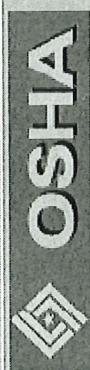
Initial Std Source Intermediate Std Source

Intermediate Std Source

Name	1280-20240716-KR
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QC Sets

Lab Number	131048	Spknt Type	Gilson 10-100uL	Serial No	UHT4270	How Many Qc Sets	4	Samples per Set	4
Last QC Sample Number	131047								



To Sol Vol

Solution - Volume

Define Solution Volume Sequence

Solution Vol1

1280-20240716-KR, Inte



Solution Vol2



Solution Vol3

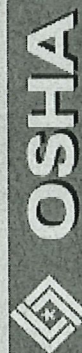
Set Sequence

lab number	Solution Vol1	Solution Vol2	Solution Vol3	Qc Smpl Seq
131048	10	0	0	1
131049	20	0	0	1
131050	40	0	0	1
131051	0	0	0	1
131052	10	0	0	2
131053	20	0	0	2
131054	40	0	0	2
131055	0	0	0	2
131056	10	0	0	3
131057	20	0	0	3
131058	40	0	0	3
131059	0	0	0	3
131060	10	0	0	4
131061	20	0	0	4
131062	40	0	0	4
131063	0	0	0	4

Preview Worksheet

Print Worksheet

Save and Exit



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KR 7/11/25

SLTC QCSM Preparation Worksheet

Input

Date: 7/11/2025
of sets to spike: 4

First QC #: 131048

Amounts Spiked (µL)

Spike 1: 10
Spike 2: 20
Spike 3: 40
Blank: 0

Analyte

Name: Fluorides (as F-)
IMIS: 1280
Synonym: Fluoride Particulates
CAS: 7664-39-3

PEL(mg/m³): 2
Rec. Air Volume(L): 90

Standard Stock

Supplier: Sigma Aldrich
Concentration/Density of spiking material: 9.023681 µg/µL
Purity: 1.000
Molecular Weight: 18.998
Lot #: MKCK9071
Standard ID: 1280-20240716-KR
Bar Code: 302325A
Date Rec'd/Prep'd: 7/16/2024
Date Expires: 7/16/2025

Media

Media Type: MCEF
Manufacturer: SKC
Catalog #: 225-1939
Lot #: 24223-7E8-064
Expiration Date: 4/30/29
Date Prepared: 7/11/25
Date Expires: 7/11/26
Prepared by: KR
Target Analytical Technique(s): SIE

General comments: Spiked using Gilson Pipetman 10-100uL, BC: UH14270

SLTC QCSM Preparation Worksheet

Spiking

Analyte: Fluorides (as F-)

CAS: [7664-39-3]

IMIS: 1280

Media: MCEF

Spiking solution:

conc/density: 9.023681 $\mu\text{g}/\mu\text{L}$

purity: 1.000

[illegible][illegible]

302325

9
KR 7/11/25**SIGMA-ALDRICH®**

sigma-aldrich.com

3050 Spruce Street, Saint Louis, MO 63103, USA

Website: www.sigmaaldrich.comEmail USA: techserv@sial.comOutside USA: eurtechserv@sial.com

Certificate of Analysis

Product Name:

Sodium fluoride – 99.99% trace metals basis

Product Number: 215309
Batch Number: MKCK9071
Brand: ALDRICH
CAS Number: 7681-49-4
MDL Number: MFCD00003524
Formula: FNa
Formula Weight: 41.99 g/mol
Quality Release Date: 16 SEP 2019

NaF

Test	Specification	Result
Appearance (Color)	White	White
Appearance (Form)	Powder	Powder
X-Ray Diffraction	Conforms to Structure	Conforms
Purity (Titration by NaOH)	≥ 98.5 %	99.1 %
Purity	Meets Requirements	Meets Requirements
99.99% Based on Trace Metals Analysis		
Trace Metal Analysis	≤ 150.0 ppm	103.8 ppm
Aluminum (Al)		4.7 ppm
Boron (B)		2.4 ppm
Barium (Ba)		0.3 ppm
Calcium (Ca)		48.0 ppm
Chromium (Cr)		0.7 ppm
Iron (Fe)		11.0 ppm
Magnesium (Mg)		2.9 ppm
Manganese (Mn)		0.8 ppm
Nickel (Ni)		0.6 ppm
Strontium (Sr)		0.3 ppm
Titanium (Ti)		0.3 ppm
Zinc (Zn)		0.3 ppm
Cesium (Cs)		7.2 ppm
Potassium (K)		2.5 ppm
Lithium (Li)		0.4 ppm
Rubidium (Rb)		0.6 ppm

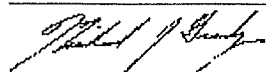
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Website: www.sigmaaldrich.comEmail USA: techserv@sial.comOutside USA: eurtechserv@sial.com

Certificate of Analysis

Product Number: 215309
Batch Number: MKCK9071



Michael Grady, Manager
Quality Control
Milwaukee, WI US

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